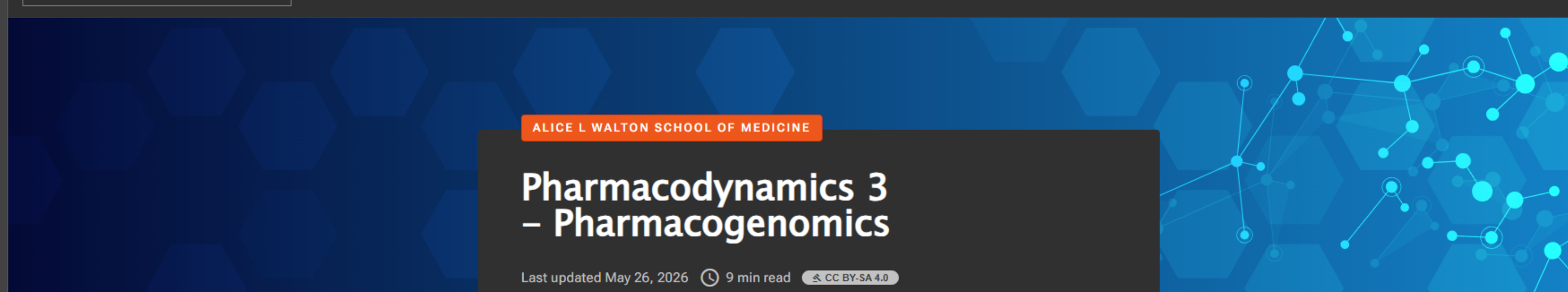

ALICE L WALTON SCHOOL OF MEDICINE

Pharmacodynamics 3 – Pharmacogenomics

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Authors: Trager Hintze, Diptiman Bose, Marie Berenguer
Editors: Jordan Finn, Russell Johnson

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Learning Objectives (3)

After completing this brick, you will be able to:

1. Predict the implications of altered pharmacokinetic or pharmacodynamic parameters in patients with genetic polymorphism.
2. Identify the key drug/ CYP enzyme relationships implicated in hepatic biotransformation.
3. Identify rele inetic principles and apply those to solve clinical problems and predict outcomes.

CASE CONNECTION

JP is a 32-year-old male who was recently diagnosed with HIV. He has no other significant past medical history. You are seeing JP in clinic for follow-up and initiation antiretroviral therapy (ART). One of the recommended regimens from an infectious disease specialist includes abacavir as part of a combination therapy. However, his provider informs him that he will need pharmacogenomic testing before starting this medication.

JP is agitated about having to have more 'lab tests'. He asks why he needs additional testing. Why does JP need pharmacogenomic testing? How would you explain the importance of pharmacogenomic testing for his particular case? Keep this in mind as you go through this brick and we will return to JP at the end.

GO TO CONCLUSION ↓

In the world of "personalized medicine," pharmacogenomics is an ever growing area. Pharmacogenomics is the science of examining inherited variations in genes that determine a patient's *response to a drug*. In fact, it is estimated that genetic factors are responsible for 20-40% of the differences in drug metabolism and response between patients. The goal of pharmacogenomics is to identify these factors and design treatments with **improved efficacy** and **reduced adverse reactions**.

How do genetic pharmacokinetics/ pharmacodynamics? (LO#1)

CYP Enzymes

Recall, from your previous genomics content, a single nucleotide polymorphism (SNP) is a change in a single nucleotide in a genetic sequence (e.g., C replaced by G). SNPs are responsible for the majority of individual variability in response to a drug.

The CYP450 enzymes are polymorphic, which means there are different forms of the same enzyme. These changes are due to a SNP in the DNA that codes for the enzyme. A SNP can cause the enzyme production in an individual to increase or decrease, which will increase or decrease the rate of drug metabolism, and, consequently, the serum level of the substrate drug.

The major types of variations are classified into 4 groups based on the effect on drug metabolism: ultra-rapid metabolizer (UM), extensive metabolizer (EM), intermediate metabolizer (IM), and poor metabolizer (PM).

- **EMs** have normal enzyme activity, which typically results in an expected drug response.
- **IMs** have fewer functional enzymes to metabolize a drug substrate. This can cause drug levels to increase (or if a prodrug is used, less will be metabolized to the active form).
- **PMs** have little or no enzyme activity, meaning they metabolize a drug more slowly, which can cause higher drug levels and increase the risk of side effects or toxicity (or, for prodrugs, less will be metabolized to the active form).
- **UMs** produce more rapid metabolism, meaning the drug is metabolized too quickly, possibly or prodrugs, more will be metabolized into the active form rapidly).

CLINICAL CORRELATION

Gene variants are more common in certain inherited and ethnic groups. Using CYP2D6 as an example, 6-10% of Caucasians are PMs and produce little or no CYP2D6, while 30% of Ethiopians are UMs and produce about double the usual amount.

HLA–B Variations

The human leukocyte antigen B (HLA-B) gene is a member of the major histocompatibility complex (MHC), and, thus, very important in coding for the immune response. In addition to affecting the peptides capable of being presented, allelic variants in the *HLA-B* gene are associated with severe, even fatal, hypersensitivity reactions for a number of medications.

While the CYP Enzyme and HLA-B variations are the most common, there are other opportunities for pharmacogenomics as well. Many of these include chemotherapy drugs. A classic example is related to the EGFR inhibitor medications (e.g., cetuximab) that binds to EGFR on tumor cells. If a patient has a particular mutation (i.e., KRAS positive) then therapy with cetuximab will NOT be effective and should not be used. This is an area that is expanding as new therapies and new variations are discovered.

Which gene variations are the most common that affect metabolism?

What are the key drug – CYP enzyme associations? (LO#2)

Recall from the "Pharmacokinetics1- 2 of 3 Drug Metabolism and Excretion" brick, that many drugs are metabolized by CYP enzymes, with CYP3A4 being the most common. You learned that drugs can be substrates of the CYP enzymes, however, they can also be *inducers* or *inhibitors* of the CYP enzymes too.

If a drug is an *inducer* it increases the rate of hepatic metabolism and decreases serum concentrations of other drugs metabolized by the same CYP enzyme. Effectively, this means that an *inducer* will cause other medications to be cleared from the body more quickly, thus making the concentration in the body less and potentially rendering them less effective.

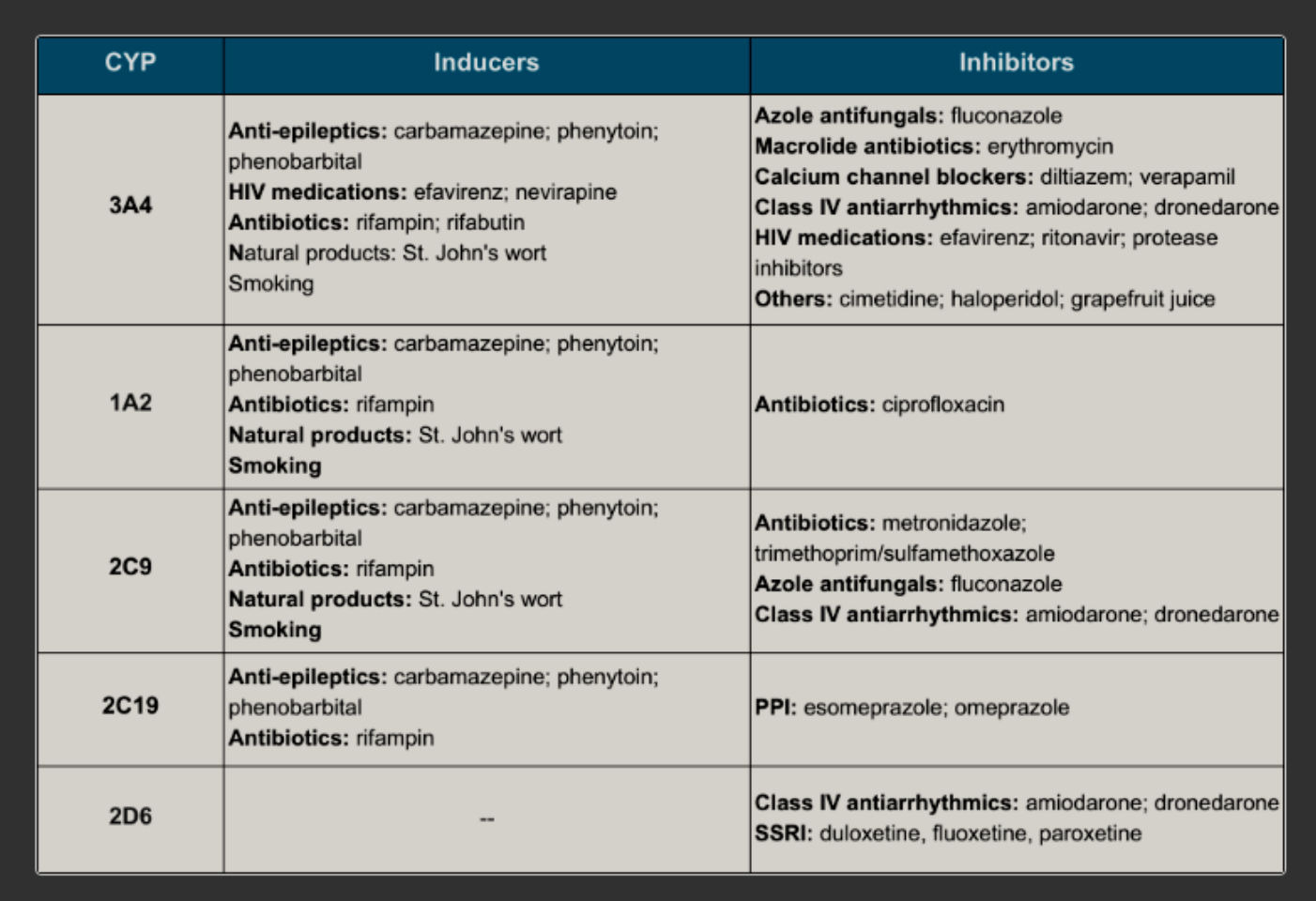
Inhibitors, on the other CYP enzyme, preventing it from in alter the rate at which drugs are metabolized, which can lead to increased or decreased drug concentrations in the body. Effectively, this means that the *inhibitor* will cause other medications to be cleared from the body more slowly, thus making concentration in the body higher and increasing the potential for toxic effects.

CLINICAL CORRELATION

CYP inhibitors can really cause a problem for patients! A specific example includes the blood thinner warfarin. It is a substrate (i.e., is metabolized by) of CYP 2C9. The antibiotic sulfamethoxazole/ trimethoprim is a strong inhibitor of that particular enzyme. So, a patient who takes warfarin and is given sulfamethoxazole/ trimethoprim for an infection (e.g., UTI) will have increased levels of warfarin in their body and, potentially, experience significant bleeding to the this drug-drug interaction.

It is important to note, drugs differ in the 'strength' to which they either induce or inhibit CYP enzymes. Medications are classified as strong, moderate, or weak inducers/ inhibitors. Drugs that are classified as 'strong' inducers/ inhibitors, of course, have the largest effect on substrate metabolism. However, patients may be taking multiple drugs that induce/ inhibit the same CYP enzyme, and, although they may be 'moderate' or 'weak', there can be an accumulation effect.

But, which drugs are most commonly associated with which CYP enzyme? Table 1 below details which drug, or which drug class, is associated with each of the most common **, and a must know!**

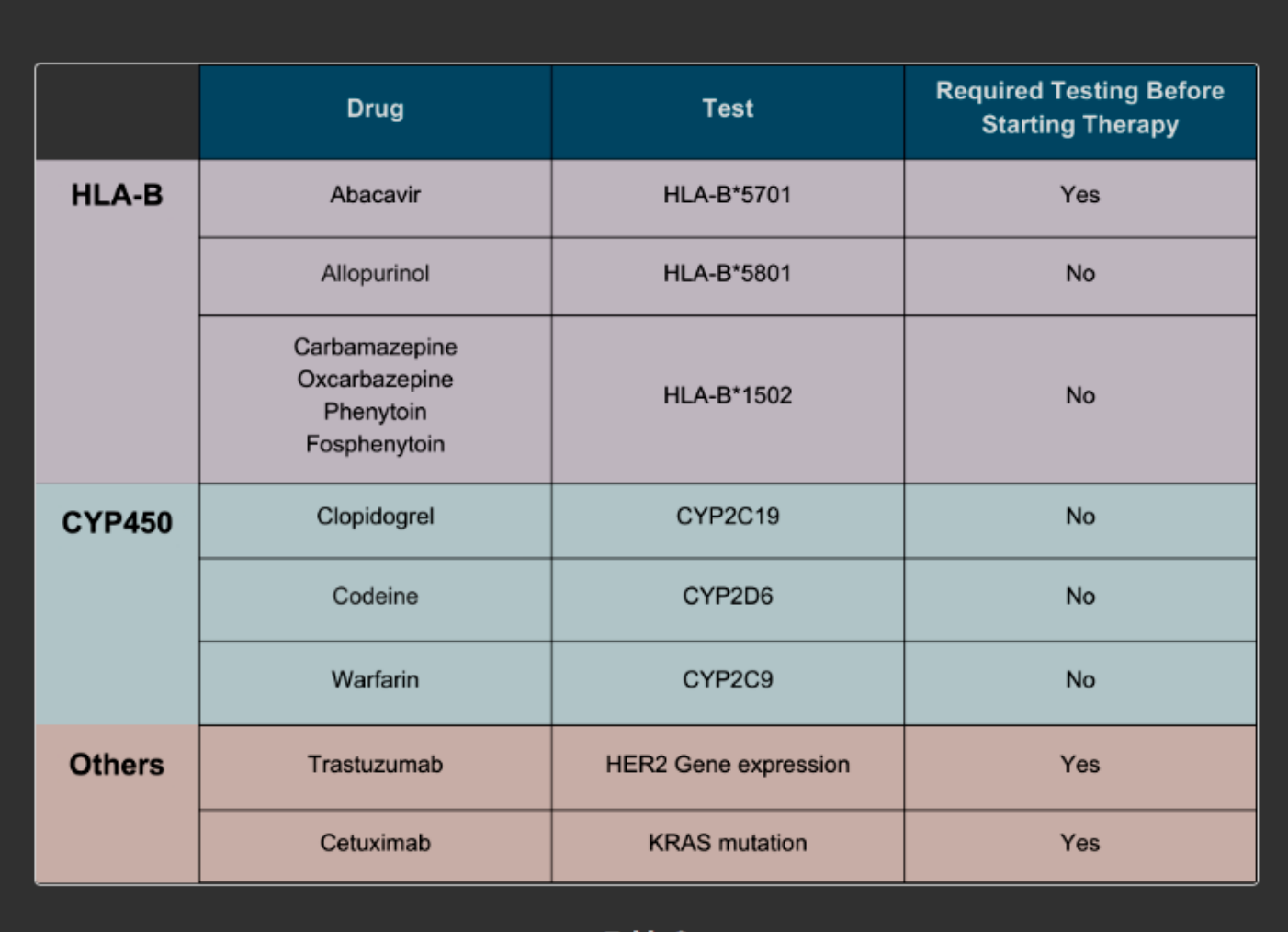


CYP	Inducers	Inhibitors
3A4	**Anti-epileptics:** carbamazepine; phenytoin; phenobarbital **HIV medications:** efavirenz; nevirapine **Antibiotics:** rifampin; rifabutin Natural products: St. John's wort Smoking	**Azole antifungals:** fluconazole **Macrolide antibiotics:** erythromycin **Calcium channel blockers:** diltiazem; verapamil **Class IV antiarrhythmics:** amiodarone; dronedarone **HIV medications:** efavirenz; ritonavir; protease inhibitors **Others:** cimetidine; haloperidol; grapefruit juice
1A2	**Anti-epileptics:** carbamazepine; phenytoin; phenobarbital **Antibiotics:** rifampin **Natural products:** St. John's wort **Smoking**	**Antibiotics:** ciprofloxacin
2C9	**Anti-epileptics:** carbamazepine; phenytoin; phenobarbital **Antibiotics:** rifampin **Natural products:** St. John's wort **Smoking**	**Antibiotics:** metronidazole; trimethoprim/sulfamethoxazole **Azole antifungals:** fluconazole **Class IV antiarrhythmics:** amiodarone; dronedarone
2C19	**Anti-epileptics:** carbamazepine; phenytoin; phenobarbital **Antibiotics:** rifampin	**PPI:** esomeprazole; omeprazole
2D6	--	**Class IV antiarrhythmics:** amiodarone; dronedarone **SSRI:** duloxetine, fluoxetine, paroxetine

Table 1

Which medications are most affected by polymorphisms? (LO#3)

There are a number of medications affected by genetic variations, and pharmacogenomic testing is becoming more common. There are some medications where testing is required prior to starting therapy, and other medications that testing may be considered later, based on response. This is a rapidly growing area, ations (**the ones you have to know!**) are listed in Table 2 below.



	Drug	Test	Required Testing Before Starting Therapy
HLA-B	Abacavir	HLA-B*5701	Yes
	Allopurinol	HLA-B*5801	No
	Carbamazepine Oxcarbazepine Phenytoin Fosphenytoin	HLA-B*1502	No
CYP450	Clopidogrel	CYP2C19	No
	Codeine	CYP2D6	No
	Warfarin	CYP2C9	No
Others	Trastuzumab	HER2 Gene expression	Yes
	Cetuximab	KRAS mutation	Yes

Table 2

ACTIVE LEARNING QUESTION

Which medication requires HLA-B*5701 testing before initiation?

A. ○ Carbamazepine
B. ● Abac
C. ○ Warfarin
D. ○ Clopidogrel
E. ○ Simvastatin

Explanation

The correct answer is Abacavir (B). HLA-B*5701 testing is required before starting Abacavir to prevent hypersensitivity reactions. Carbamazepine (A) requires HLA-B*1502 testing in certain populations. Warfarin (C) involves CYP2C9 and VKORC1 testing. Clopidogrel (D) involves CYP2C19 testing. Simvastatin (E) involves SLCO1B1 testing.

CASE CONNECTION

BACK TO INTRODUCTION ↑

Returning to JP. As you have learned, there are certain medications that require pharmacogenomic testing prior to their initiation. This includes the HIV medication abacavir. Patients who test **positive for HLA-B*5701 should NOT receive abacavir** due to a significantly increased risk of life-threatening hypersensitivity reactions.

As you express concern and for JP, and empathize with him about his new diagnosis, you would ing is needed in order to avoid poten this additional lab test, to see if he has the mutation, it will help us determine the most safe and effective therapy for him going forward.

Summary

- Pharmacogenomics studies genetic variations that affect individual responses to drugs, aiming to improve treatment efficacy and reduce adverse reactions.
- Single nucleotide polymorphisms (SNPs) in CYP450 enzymes significantly influence drug metabolism, classifying individuals as ultra-rapid, extensive, intermediate, or poor metabolizers.
- HLA-B gene variations can lead to severe hypersensitivity reactions to certain medications, highlighting the importance of genetic testing before drug administration.
- CYP enzymes can be induced or inhibited by drugs, affecting the metabolism and serum concentration of other drugs metabolized by the same enzyme.
- Pharmacogenomic testing is crucial for certain medications, such as abacavir and trastuzumab, to ensure effective and safe treatment.
- Understanding drug-CYP enzyme associations is essential for predicting drug interactions and optimizing therapeutic outcomes.

Review Questions

1. Which group of ind xicity due to slow drug metabolism?

A. ○ Ultra-rapid metabolizers
B. ○ Extensive metabolizers
C. ● Poor metabolizers
D. ○ Intermediate metabolizers
E. ○ Ethnicity

Hide Explanation

The correct answer is poor metabolizers (C). Poor metabolizers (PMs) have little or no enzyme activity, meaning they metabolize a drug more slowly, which can cause higher drug levels and increase the risk of side effects or toxicity. Ultra-rapid metabolizers (A) metabolize drugs too quickly, possibly leading to subtherapeutic levels. Extensive metabolizers (B) have normal enzyme activity, which typically results in an expected drug response. Intermediate metabolizers (D) have fewer functional enzymes to metabolize a drug substrate. This can cause drug levels to increase, but not to the same extent as in poor metabolizers. Ethnicity (E) can influence the prevalence of certain genetic polymorphisms, but it is not a direct determinant of drug metabolism speed.

2. Which CYP enzyme is most commonly associated with the metabolism of warfarin?

A. ○ CYP3A4
B. ● CYP2C9
C. ○ CYP2D6
D. ○ CYP1A2
E. ○ CYP2C19

Hide Explanation

The correct answer is CYP2C9 (B). Warfarin is primarily metabolized by the CYP2C9 enzyme. CYP3A4 (A) is involved in the metabolism of many drugs, but not warfarin. CYP2D6 (C) is known for metabolizing drugs like codeine and antidepressants. CYP1A2 (D) is involved in the metabolism of drugs like caffeine. CYP2C19 (E) metabolizes drugs such as clopidogrel.

3. Which medication requires HLA-B*5701 testing before initiation?

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How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Predict the implications of altered pharmacokinetic or pharmacodynamic parameters in patients with genetic polymorphism.~~
02 ☑ ~~Identify the key drug/ CYP enzyme relationships implicated in hepatic biotransformation.~~
03 ☑ ~~Identify kinetic principles ems and predict~~

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02 ☑ ~~Identify the key drug/ CYP enzyme relationships implicated in hepatic biotransformation.~~
03 ☑ ~~Identify relevant pharmacodynamic and pharmacokinetic principles and apply those to solve clinical problems and predict outcomes.~~

Objective Confidence: 3/3